

Grade 6
Unit 11
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Determine if each pair of expressions are equivalent.

1. $-2(10 + 4)$ and $-2(10) + 2(4)$
 $-2(14)$ $-20 + 8$
 -28 $\neq$ -12
 not equivalent

2. $0.5(5 - 2)$ and $0.5 \cdot 5 + 0.5 \cdot -2$
 $= 0.5(3)$ $= 2.5 - 1$
 1.5 $=$ 1.5
 equivalent

3. $\frac{3}{4} \times \frac{9}{10}$ and $\frac{9}{10} \times \frac{3}{4}$
 $= \frac{3}{4} \times \frac{9}{10}$ $= \frac{9}{10} \times \frac{3}{4}$
 $= \frac{27}{40}$ $= \frac{27}{40}$
 equivalent

4. $\left(\frac{2}{3} + \frac{1}{8}\right) + \frac{1}{6}$ and $\frac{2}{3} + \left(\frac{1}{8} + \frac{1}{6}\right)$
 $8\left(\frac{2}{3}\right) + \left(\frac{1}{8}\right)^3 + \left(\frac{1}{6}\right)^4$ $8\left(\frac{2}{3}\right) + \left(\frac{1}{8}\right)^3 + \left(\frac{1}{6}\right)^4$
 $\left(\frac{16}{24} + \frac{3}{24}\right) + \frac{4}{24}$ $\frac{16}{24} + \left(\frac{3}{24} + \frac{4}{24}\right)$
 $\frac{19}{24} + \frac{4}{24} = \frac{23}{24}$ $\frac{16}{24} + \left(\frac{7}{24}\right) = \frac{23}{24}$
 equivalent

5. $\frac{1}{6} + \frac{2}{9}$ and $\frac{2}{9} + \frac{1}{6}$
 $\frac{3}{3}\left(\frac{1}{6}\right) + \left(\frac{2}{9}\right)^2$ $2\left(\frac{2}{9}\right) + \left(\frac{1}{6}\right)^3$
 $\frac{3}{18} + \frac{4}{18}$ $\frac{4}{18} + \frac{3}{18}$
 $\frac{7}{18}$ $=$ $\frac{7}{18}$
 equivalent

6. $\frac{7}{11} - \frac{1}{3}$ and $\frac{1}{3} - \frac{7}{11}$
 $\frac{3}{3}\left(\frac{7}{11}\right) - \left(\frac{1}{3}\right)^{\frac{11}{11}}$ $\frac{11}{11}\left(\frac{1}{3}\right) - \left(\frac{7}{11}\right)^{\frac{3}{3}}$
 $\frac{21}{33} - \frac{11}{33}$ $\frac{11}{33} - \frac{21}{33}$
 $\frac{10}{33} \neq \frac{-10}{33}$
 not equivalent

7. $\frac{5}{7} + \left(\frac{1}{2} + \frac{3}{4}\right)$ and $\left(\frac{5}{7} + \frac{1}{2}\right) + \frac{3}{4}$
 $4\left(\frac{5}{7}\right) + \frac{14}{14}\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right)^{\frac{7}{7}}$ $\left(\frac{20}{28} + \frac{14}{28}\right) + \frac{21}{28} = \frac{55}{28}$
 $\frac{20}{28} + \left(\frac{14}{28} + \frac{21}{28}\right) = \frac{20}{28} + \frac{35}{28} = \frac{55}{28}$
 equivalent

8. $\left(\frac{1}{5} - \frac{2}{5}\right) - \frac{8}{9}$ and $\frac{1}{5} - \left(\frac{2}{5} - \frac{8}{9}\right)$
 $\left(\frac{1}{5} - \frac{2}{5}\right) - \frac{8}{9}$ $\frac{1}{5} - \left(\frac{2}{5} - \frac{8}{9}\right)$
 $-\frac{1}{5} - \frac{8}{9}$ $\frac{1}{5} - \left(\frac{18}{45} - \frac{40}{45}\right)$
 $-\frac{9}{45} - \frac{40}{45} = \frac{-49}{45}$ $\frac{1}{5} - \left(-\frac{22}{45}\right)$
 $\frac{9}{45} + \frac{22}{45} = \frac{31}{45}$
 not equivalent

9. Rewrite $(6x - 2y) + 9$ using the associative property.
 $6x + (-2y + 9)$

10. Rewrite $9z - 7b$ using the commutative property.
 $-7b + 9z$